Forest Sensor Sprint

Site the smartest place for a campus tree sensor

At a glance: Find the smartest place to install a future campus tree sensor. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: tree height, microclimate, temperature, humidity, light, evidence-based site selection (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 clipboards (one per team) plus 1 spare
- 4 paper clinometers (one per team) plus 2 spares
- 4 combined temperature and humidity meters (one per team) plus 1 spare; one combined meter, not two device types
- 4 three-in-one soil meters (moisture, light, pH), one per team, plus 1 spare; the light position replaces a separate light meter
- 1 long tape measure to pre-mark the tree-height standoff distance
- 4 route cards (one per team) plus 2 spares
- pencils, shared

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

03 No PPE: this is an outdoor measurement walk with no chemicals, water, or soil handling. Set up sun and hydration support, run the weather/heat-index check, confirm the buddy system and route boundaries, and pre-mark the tree-height standoff distance with the long tape.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook
0:00 - 0:05

Pose the mission as a challenge: "Find the smartest place to install a future campus tree sensor." Take a quick prediction by show of hands.

Phase 2 Prediction
0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints
0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve
0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Measure and record

Teams walk the checkpoints and record temperature, humidity, the soil-meter light position, and soil moisture with units at each, and estimate the marked tree's height from the pre-marked standoff distance.

Phase 6 Route and recommend

Teams order their checkpoints to minimize backtracking, then pick one sensor site and defend it with their own numbers.

Phase 7 Score, debrief, cleanup
last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Wipe the soil-meter probes and return all instruments to the bin; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: A meter reads the same everywhere. Fix: Check the soil meter is on the right switch position and the probe is fully in the soil, give the temperature/humidity meter time to settle, and compare a shaded spot against open lawn so the microclimate difference shows.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to microclimate varies in meters.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	**100**

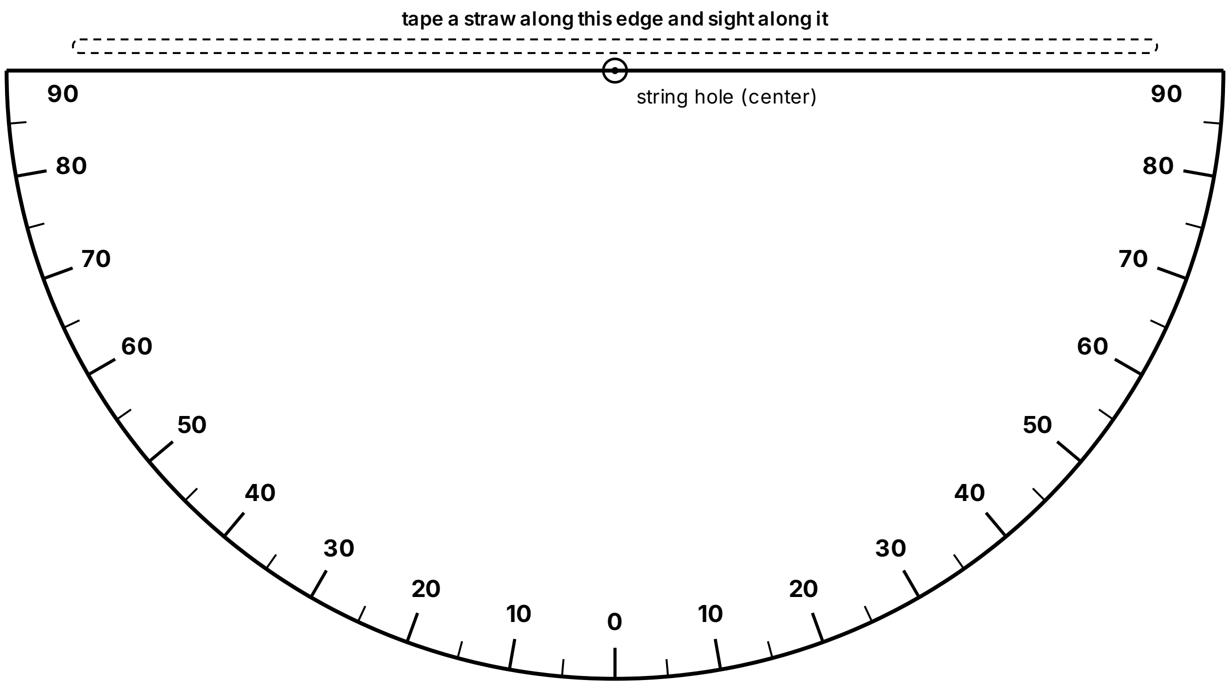
CLEANUP AND SOURCES

Wipe the soil-meter probes, return clipboards, meters, clinometers, and the tape to the bin, and collect route cards and score slips. The only consumables are the printed clinometers and route cards.

SOURCES GLOBE Observer Trees; NASA MyNASAData Urban Heat Island

Print on cardstock for TTT-02 Forest Sensor Sprint. Four teams plus spares. Clinometer: one per page, print 6 (4 teams + 2 spares). Route cards: two per page, print 3 sheets for 6 cards (4 teams + 2 spares).

Paper clinometer



Cut along the flat top edge and the curved outline.

ASSEMBLE

01 Cut out the half circle along the curved outline.

02 Tape a straight straw along the top straight edge.

03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.

04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.

05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor marked.

ANGLE	TAN
10	0.18
20	0.36
30	0.58
40	0.84
45	1.00
50	1.19
60	1.73
70	2.75

Example: eye height 1.5 m, distance 10 m, angle 40° gives 1.5 + 10 × 0.84 = about 9.9 m.

Team route card

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ________________________   Date ______________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: ____

Why (use your data): ____

The checkpoints are the spots the instructor marked outside.

cut here

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

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